

# Juan Pablo Espinosa Chessal

Robotics Engineer · Research lead of GR-0X Humanoid Robot

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## WORK EXPERIENCE

**Founder & Research Lead — GR-0X Humanoid, ALMaS Research Group (WPI)** May 2025 – Present

- **Founded and lead a 12-person research team** (3 M.S. students, 5 seniors) through WPI's Major Qualifying Project (MQP) to build **GR-0X** — a single humanoid platform that reconfigures its own morphology to take on the advantages of multiple robot body types for different tasks — protected under a **patented morphology-aware system (USPTO 63/963896)**.
- **Secured sponsorship and in-kind support** from **Microchip Technology, Polymaker, and G20 Inc.**, funding hardware, materials, and compute for the project.
- **Fully validated (100%) the adaptive lower-body/leg subsystem** (patent pending), with **working RL-based locomotion policies**; now developing the upper-body subsystem (torso, arms, actuation) and preparing sim-to-real transfer.
- **Architected a distributed multi-brain control system** (ROS 2) and directed full mechanical and electrical integration across the platform.
- **Advise 2 of 3 M.S. thesis projects** within the lab, guiding research direction on morphology-performance trade-offs (strength, reach, speed).

**Embedded Engineer Intern, Inbioidroid — León, Mexico** Dec 2024 – Jun 2025

- Contributed to **Prometheus 3.0**, Mexico's first avatar humanoid robot, at a finalist company in the **ANA Avatar XPRIZE**.
- **Designed and integrated a LiDAR-based SLAM system** achieving **<5 cm mapping accuracy** for autonomous navigation.
- Presented Prometheus 3.0 to **government officials** as the youngest member of a 10-person engineering team.
- **Co-inventor** on a pending Mexican patent for a telepresence control system that **improved response latency by ~20%**.

## PROJECTS

**FIRST Robotics Competition – Teams 3480 & 3284** Aug 2023 – May 2025

- Directed the **programming and electrical division** for an award-winning FRC team.
- Developed autonomous routines using **path planning, computer vision, and odometry**.
- **Laguna Regional Champion**, qualifying for the **2024 World Championship**.

**AI Vision Sumo Robot – León, Mexico** Nov 2024

- Built an autonomous robot using **YOLO-based vision** with **sub-100 ms response time**.

## SKILLS

|                       |                                                                                                                                                    |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>    | C++, Python, Bash, Java, C, C#, Git, PyTorch, SLAM, ML, RL                                                                                         |
| <b>Robotics</b>       | Embedded Systems, Sensor Integration, Control Systems, Locomotion, Computer Vision, RL Policy Training, Domain Randomization, Sim-to-Real Transfer |
| <b>Mechanical</b>     | CAD (SolidWorks, Onshape, Inventor), 3D Printing, Manufacturing, Material Analysis & Simulation                                                    |
| <b>Electronics</b>    | PCB Design, Sensors/Actuators, Circuit Debugging, Microcontrollers                                                                                 |
| <b>Software Tools</b> | Isaac Sim, Isaac Lab, Isaac Gym, ROS 2, Gazebo, RSL-RL, Linux, MATLAB, Overleaf                                                                    |
| <b>Languages</b>      | Spanish (Native), English (Fluent), German (Basic)                                                                                                 |

## EDUCATION

**Worcester Polytechnic Institute (WPI)** May 2029

*B.S. in Robotics Engineering and Computer Science* GPA: **3.71/4.0**

- **Charles O. Thompson Scholar** (top first-year academic honor)
- **Presidential Scholarship** and **International Grant**

**Instituto Tecnológico de Monterrey: PrepaTec** Aug 2023 – May 2025

*High School Diploma* GPA: **95.7/100**

- **Merit Scholarship** and **Honor Mention at Graduation**

## ACTIVITIES & LEADERSHIP

**Conference Chair (Executive Board), IEEE Student Branch — WPI** Aug 2025 – Present

**President & Founder, International Club of Unity (I.C.U.)** Aug 2024 – May 2025

**Vice President, Student Government (SEPREPA)** Aug 2024 – May 2025